FACULTY OF COMPUTING

SEMESTER 1 2024/2025

LAB 4 - ARP and Switch Table

SECR 1213 – NETWORK COMMUNICATIONS

SECTION 12

LECTURER: MR. FIROZ BIN YUSUF PATEL DAWOODI

NAME	MATRIC NUMBER
NUR FIRZANA BINTI BADRUS HISHAM	A23CS0156

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

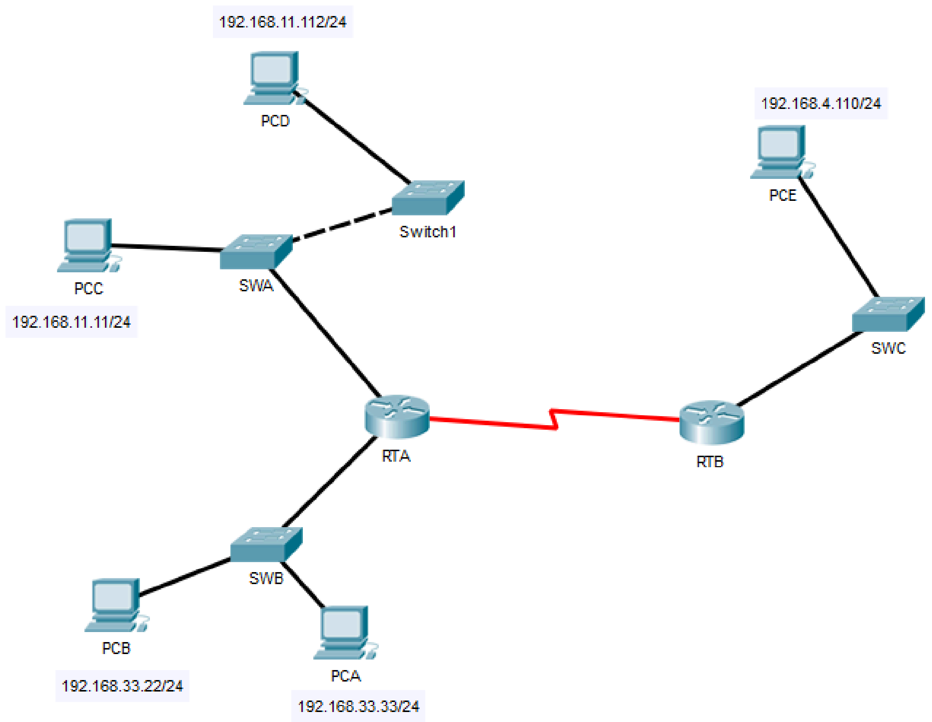


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.11.1             -  0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.33.1             -  000C.CF0C.593A  ARPA   FastEthernet0/0
```

RTA Router

b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.4.1              -  0001.977A.B614  ARPA   FastEthernet0/0
```

RTB Router

c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0002.4a00.0e91    DYNAMIC     Fa0/1
   1    000c.8546.7d85    DYNAMIC     Fa1/1
```

SWA switch

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.cf0c.593a    DYNAMIC     Fa0/1
```

SWB Switch

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.977a.b614    DYNAMIC     Fa0/1
```

SWC Switch

d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp –a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCA

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCB

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCC

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

e. What are your thoughts on the results?

The results show there is no entry for all PCs because all of the PCs didn’t receive any ARP request yet. There will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks to reduce the amount of network traffic viewed in the simulation.

a. Click **PCA** and click the Desktop tab > Command Prompt.

b. Enter the **ping 192.168.33.22** command. This may take a few seconds.

c. In the Command prompt of PCA, type **arp –a**. Paste the result of this command here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time=2ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.33.22         0060.47ea.a746        dynamic
```

PCA

d. In the Command prompt of PCB, type **arp –a**. Paste the result of this command here

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.33.33         0002.1755.9a06        dynamic

C:\>
```

PCB

e. In the Command prompt of PCC, PCD and PCE, type **arp –a**. Paste the result of this command here.

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCC

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

Step 2: Generate traffic between PCC to all other PCs except PCA.

a. Click **PCC** and click the Desktop tab > Command Prompt.

b. Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp –a**. Paste the result after these commands here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time=2ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.11.1          0002.4a00.0e91        dynamic
```

(Ping to PCB)

c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp –a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.11.1          0002.4a00.0e91        dynamic
  192.168.11.112        0001.6462.0278        dynamic
```

(Ping to PCD)

d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp –a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=2ms TTL=126
Reply from 192.168.4.110: bytes=32 time=3ms TTL=126
Reply from 192.168.4.110: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 12ms, Average = 5ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.11.1          0002.4a00.0e91        dynamic
  192.168.11.112        0001.6462.0278        dynamic
```

(Ping to PCE)

e. Discuss the results you got from all the commands on PCC.

The ARP table on PCC remains empty until it pings to PCB, PCD and PCE. No ARP entries are present to these ping operations. The ARP table moves into only after PCB, PCD and PCE receive ARP requests from PCC and respond to the subsequent ping messages.

f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

RTA>***enable*** RTA#***show arp***

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.11.1            -   0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11           6   00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1            -   000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22           6   0060.47EA.A746  ARPA   FastEthernet0/0
RTA#
```

RTA Router

g. At Router RTB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

RTB>***enable*** RTB#***show arp***

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.4.1             -   0001.977A.B614  ARPA   FastEthernet0/0
Internet  192.168.4.110           5   0060.702D.7C08  ARPA   FastEthernet0/0
```

RTB Router

Step 3: Switch MAC address table.

a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

SWA>***enable*** SWA#***show arp*** SWA#***show mac-address-table***

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0002.4a00.0e91    DYNAMIC     Fa0/1
   1    000c.8546.7d85    DYNAMIC     Fa1/1
```

SWA Switch

b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

SWB>***enable*** SWB#***show arp*** SWB#***show mac-address-table***

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.cf0c.593a    DYNAMIC     Fa0/1
```

SWB Switch

c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

SWC>***enable*** SWC#***show arp*** SWC#***show mac-address-table***

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.977a.b614    DYNAMIC     Fa0/1
SWC#
```

SWC Switch

```
Switch>enable
Switch#show arp

Switch#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0005.5e95.187b    DYNAMIC     Fa0/1
Switch#
```

Switch 1

d. Do switches use an arp table? (Y/N)

   Yes.

e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetCom syllabi, it is just for knowledge.*

   Multilayer switches use ARP tables because they need to resolve IP addresses into MAC addresses to route packets between subnets or VLANs. This functionality bridges the gap between Layer 2 switching and Layer 3 routing, making ARP tables essential for efficient and accurate packet delivery.

f. What information does the command ***show mac-address-table*** give?

   To view the MAC address table entries on a switch, the DYNAMIC refers to MAC addresses that the switch has automatically learned during its operation.

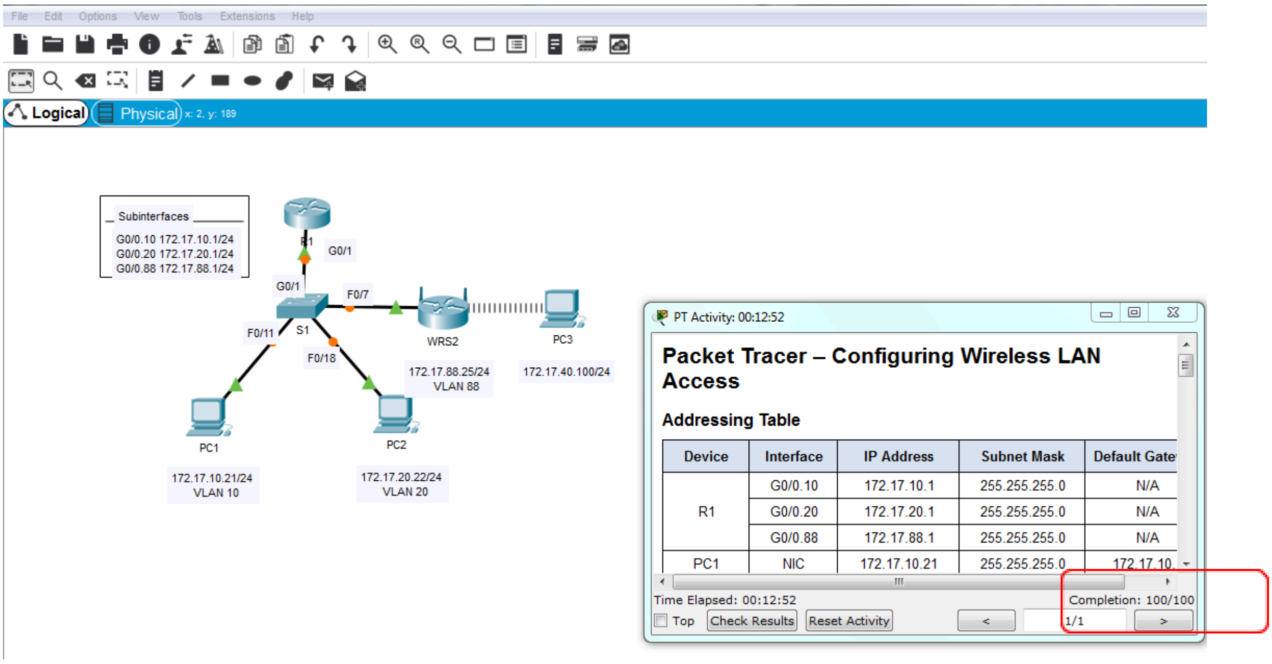
Part 3: Attach wireless lab results.

In this part, you will use the Lab 4 .pka file.

Step1: Change the filename of Lab 4.

a. Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*

b. Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.



c. Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.